



Manifold for Type 1 Diabetes – Restoration

NIDDK Human Islet Research Network — cross-consortium multi-omic causal integration

1 · THE CLINICAL PROBLEM

The NIH NIDDK Human Islet Research Network (HIRN) was established on the recognition that T1D restoration research is fundamentally a multi-disciplinary integration problem. Its four operating consortia — CBDS (Beta-Cell Death and Survival), CTAR (Targeting and Regeneration), CHIB (Human Islet Biomimetics and Bioengineering), and HIPPI (Human Islet Phenotyping Program) — each contribute rich, orthogonal signal to the same underlying biological substrate. The field-level analytic gap is the integration itself:

- **Cross-consortium causal integration.** β -cell death pathways (CBDS), regenerative target space (CTAR), engineered-islet biomimetics (CHIB), and phenotypic characterization (HIPPI) cohere biologically but are typically analyzed within consortium boundaries. The directed structure connecting them — which death cascade responds to which regenerative intervention in which engineered substrate under which phenotypic profile — is the natural domain of a causal-discovery engine.
- **Multi-omic data volume has outrun conventional integration.** HIPPI's deep phenotyping of human islets, combined with the cross-consortium transcriptomic, proteomic, and functional-assay output, exceeds what parallel-correlation GWAS-style integration recovers. Principled conditional-independence-based directed-acyclic-graph inference is the tool the volume warrants.
- **Translational pipeline requires interdiction framing.** Moving HIRN-generated biology into trial-ready interventions — whether pharmacologic (CTAR), engineered (CHIB), or cell-therapy-adjacent — is a rational-ordering problem on the inferred causal graph, not a discovery problem.

Three capabilities remain underdeveloped across the HIRN analytic surface:

- Unified causal modeling across consortia — integrating β -cell death cascade (CBDS), regenerative-target pathway (CTAR), biomimetic-engineering substrate (CHIB), and phenotypic characterization (HIPP) into a single directed graph rather than parallel consortium analyses.
- Principled target prioritization across the combined pipeline — which regenerative target attenuates which death cascade under which engineered context, ranked by composite intervention leverage.
- Composite-fragility framing for cross-consortium benchmarking — a standardized ΩF decomposition that allows CBDS death markers, CTAR regenerative readouts, CHIB graft-performance measures, and HIPP phenotype signatures to be compared and integrated on a common analytic substrate.

HIRN's constituent consortia are individually world-class. The field-level question is whether their combined output can be integrated on a single causal substrate to produce translational leverage that exceeds the sum of consortium outputs — a problem of cross-consortium causal inference and principled interdiction.

2 · WHAT MANIFOLD BRINGS

Manifold is a causal-intelligence platform for detecting structural fragility and regime transitions in complex multi-signal systems. Three engine capabilities form the basis of the HIRN-configured deployment:

1. **Causal discovery (Spirtes / PC algorithm) on cross-consortium multi-omic data.** Recovers the directed structure among death-cascade markers (CBDS), regenerative-pathway targets (CTAR), engineered-substrate performance (CHIB), and phenotype signatures (HIPP), providing the cross-consortium causal substrate.
2. **Criticality estimators feeding a composite fragility score (ΩF).** Configured as a unifying benchmark across consortium outputs — heterogeneous-treatment-effect learners on CTAR intervention data, persistent-homology and topological estimators on HIPP phenotypic manifolds, change-point estimators on CHIB graft-performance trajectories, survival models on CBDS death-cascade readouts — with the composite score providing a stable integration layer.
3. **Pearl do-calculus and interdiction on the cross-consortium graph.** Translational-pipeline prioritization — which regenerative target, engineered substrate, and phenotypic patient subgroup to advance — is framed as maximum-attenuation intervention on the inferred directed structure, producing principled target rankings rather than consortium-by-consortium empirical selection.

3 · OF PILLAR MAPPING - HIRN CROSS-CONSORTIUM PHYSIOLOGY

OF PILLAR	HIRN READING
I — Interaction	Cross-consortium signal interactions: death-cascade × regenerative-target × engineered-substrate × phenotype coupling
R — Reserve	Baseline and engineered β -cell functional mass across HIPPP phenotyping and CHIB biomimetic substrates
J — Junction	Immune-metabolic and death-regeneration cascade junctions; cross-consortium transfer-entropy structure
C — Cascade	CBDS death cascades (ER stress, apoptosis, autophagy) and their interaction with CTAR regenerative cascades (DYRK1A, mTOR, FOXO1)
T — Temporal	Hazard of β -cell loss, graft compromise, or regenerative response loss across the integrated pipeline

4 · CRITICALITY ESTIMATOR CONFIGURATION

HIRN data is heterogeneous across consortia in modality, sample size, and longitudinal structure. The criticality layer is configured to accommodate the full range:

ESTIMATOR	WHAT IT CATCHES	PILLAR FEED
Persistent homology on HIPP multi-omic phenotypic manifolds	Topological regime detection across donor phenotypes; identification of discontinuous phenotypic clusters	I, C
Heterogeneous-treatment-effect learners (Causal Forest, BART) on CTAR intervention data	Which targets produce differential response across HIPP-characterized phenotypic subgroups	I, C
Bayesian online change-point (BOCPD) on CHIB graft-performance trajectories	Graft-compromise and biomimetic-substrate failure detection with calibrated uncertainty	J, T
Survival and mixed-effects models on CBDS death-cascade markers	Per-donor death-cascade hazard; time-to-apoptosis given stressor exposure	R, T

Additional estimators available within the same layer for subsequent phases:

- **Transfer entropy** across consortium signal modalities — quantifies directional coupling between CBDS death-cascade activation, CTAR regenerative-pathway engagement, CHIB substrate response, and HIPP phenotypic state.
- **Graph neural networks** on the discovered cross-consortium causal graph — provides tractable inference under interventional queries once the graph is stabilized.

5 · PROPOSED INITIAL ENGAGEMENT (PHASE 1)

The scope of the proposed Phase 1 engagement is defined to produce a deliverable of cross-consortium analytic value within a contained time and data footprint.

- **Primary dataset.** HIRN-shared multi-omic and functional data across at least two consortia — candidates include HIPPI human-islet phenotyping output paired with either CBDS death-cascade assays or CTAR regenerative-intervention data. Cross-consortium integration is the analytic target rather than any single-consortium deepening.
- **Access path.** HIRN coordinating-center-sponsored Data Use Agreement, under the co-sponsorship of consortium principal investigators across the selected analytic pairing.
- **Deliverable.** Cross-consortium Ω F decomposition integrating at least two consortium output streams onto a single causal graph. Prioritized target ranking via Pearl do-calculus with interdiction. Reproducible analytic pipeline suitable for community release as a HIRN-sanctioned methodological deliverable.
- **Success criteria.** Cross-consortium causal graph whose structure is validated against pre-specified biological priors from consortium PIs; prioritized target ranking that identifies at least one HIRN-tractable intervention pathway not prominent in single-consortium analyses; methodological deliverable suitable for publication in a HIRN-affiliated venue.
- **Timeline.** Twelve to sixteen weeks from data access, reflecting the cross-consortium integration complexity.
- **Excluded from Phase 1** (reserved for the Phase 2 expansion): integration across all four HIRN consortia simultaneously, prospective experimental validation, and bespoke assay development.

6 · RATIONALE AND TIMING

- **HIRN is structurally positioned for integrative analytics.** The consortium architecture was established explicitly to produce cross-modality integration; the analytic tools capable of delivering that integration at scale — causal-discovery engines on multi-omic data, composite-fragility decomposition — have only recently matured to production grade.
- **Replacement and regeneration paradigms are converging.** Insulin-independence in early-phase Vertex VX-880 cohorts and reproducible human β -cell proliferation via DYRK1A inhibition both reinforce that the near-term bottleneck is not biological discovery but principled prioritization across modalities — the problem HIRN was built to address.
- **Cross-consortium prioritization has direct trial-design consequences.** The targets and substrates HIRN advances become the candidate interventions for subsequent clinical studies. Principled interdiction-based ranking reduces the combinatorial design space actionably.
- **Funding mechanisms are aligned.** The NIH NIDDK HIRN itself, the Breakthrough T1D Cures program, and the Helmsley Charitable Trust Restoring Insulin Production stream all support cross-modality integrative analytics as a defined funding priority.
- **The platform is production-ready.** Manifold's causal, criticality, and interdiction engines are production-grade; cross-consortium configuration represents bounded, well-scoped integration work.

7 · PROPOSED PARTNERSHIP

Partnership with HIRN coordinating-center leadership and co-sponsoring consortium principal investigators is sought for:

1. Cross-consortium retrospective data access under the HIRN coordinating-center Data Use Agreement framework.
2. Co-authorship on the cross-consortium causal-integration methodological manuscript, positioned as a HIRN-affiliated deliverable.
3. Co-application to NIH NIDDK HIRN expansion, Breakthrough T1D Cures, or the Helmsley Charitable Trust for the Phase 2 arm extending integration across additional consortia.

In exchange, Apex Analytica contributes the causal-inference engine, the Ω F framework, and the engineering team. Apex retains commercial rights to the platform; research intellectual property from the joint work is shared on standard academic-industry terms.